

Arshad Khatib

+91-8806045005 | arshadkhatib.2006@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#) | [Codeforces](#)

SUMMARY

AI/ML Engineer specializing in deep learning, natural language processing, and production-grade machine learning systems. Three-time national hackathon winner, published researcher (Kaggle global rank 4), with hands-on expertise in PyTorch, LLMs, MLOps, agentic AI workflows, and low-latency model deployment at scale.

EDUCATION

•Sardar Vallabhbhai National Institute of Technology (SVNIT)

2024 – Present

Bachelor of Technology in Artificial Intelligence (Institute of National Importance) | CGPA: 7.99/10

Surat, India

– **Relevant Coursework:** Machine Learning, Data Structures, Design and Analysis of Algorithms, Linear Algebra, Probability and Statistics, Operating Systems, Computer Networks, Database Management Systems, Theory of Computation.

TECHNICAL SKILLS

Machine Learning and AI: PyTorch, TensorFlow, Scikit-Learn, XGBoost, LangChain, RAG, Transformers (BERT, ViT), LLMs, Deep Learning, NLP, Computer Vision, Time Series Forecasting, Anomaly Detection, Feature Engineering, MLOps, Model Context Protocol (MCP)

Languages: Python, C/C++, SQL, R, Bash/Shell, JavaScript, TypeScript

Web and Deployment: FastAPI, Next.js, Docker, Node.js, AWS (SageMaker, EC2, ECS, RDS, S3), Google Cloud Platform, REST APIs, gRPC

Tools and Libraries: Git, OpenCV, Pandas, NumPy, HuggingFace, Postman, LaTeX, TimescaleDB

EXPERIENCE

•Machine Learning Researcher – Astrophysical Document Classification

September 2025 – October 2025

TRACS @ WASP 2025 Shared Task | Team: *Clutch or Cry*

[Published Paper](#)

- **Ranked 4th globally** (Macro F1: **0.82**) by building a two-level stacking ensemble: a 1,000+ keyword rule-based classifier and fine-tuned AstroBERT (trained on 440K astrophysical abstracts) fused into a 7-dimensional meta-feature vector for four independent XGBoost meta-learners.
- **Resolved extreme class imbalance** (1:91 instrumentation ratio) using SMOTE oversampling, class-weighted training (scale_pos_weight = 180), and calibrated decision thresholds; lifted INSTRUMENTATION F1 from **0.510 to 0.782** (+53.3%) without degrading majority-class performance.
- **Delivered 97% accuracy on Task 1** with a hybrid Random Forest and rule-based bibcode heuristic, iterated over 5 architectures (LSTM, AstroBERT, Logistic Regression, Random Forest, final ensemble); methodology documented in a peer-reviewed publication.

PROJECTS

•Prometheus – Intelligent Grid Operations and Optimization

May 2026

1st Place Winner, PowerMind Hackathon (SVNIT x Adani)

[GitHub](#)

- **Engineered** a grid intelligence forecasting system achieving **0.59% MAPE** and $R^2 = 0.998$ via a 6-model ensemble (LightGBM, XGBoost, CatBoost, Prophet, TCN, stacking), producing **24-hour, 7-day, and 30-day** demand forecasts with P10/P50/P90 quantile uncertainty bands.
- **Developed** an agentic AI layer integrating **7 MCP tools** (Ollama qwen2.5:3b) for real-time load balancing, equipment scheduling, and IEX surplus energy optimization, reducing manual operator intervention by 40%.
- **Implemented** an end-to-end MLOps pipeline featuring seasonal z-score anomaly detection, LLM-powered event explanations, a versioned model registry, and automated PDF reports covering EBITDA and carbon credit metrics.

•HFT Market Analysis and Trading Platform

December 2025 – January 2026

1st Place Winner, Genesis 2026 | Surveillance and Pattern Discovery

[GitHub](#)

- **Architected** an HFT surveillance platform ingesting **160+ Binance order book snapshots/sec**, extracting **40+ microstructure features** (OFI, VPIN, OBI); a DeepLOB CNN model achieved **63.4% directional accuracy** via 5-fold cross-validation.
- **Deployed** C++ gRPC microservices at **0.5 ms average latency** (4.4× faster than Python baseline); the paper trading engine achieved a **59.6% win rate**, Sharpe ratio of **1.82**, and \$287 realized PnL across 94 simulated trades.
- **Automated** spoofing and layering detection across **1.3M+ stored snapshots** in TimescaleDB (8:1 compression, 42 ms query time); deployed on AWS (ECS, RDS, S3) with 87% test coverage across 95+ unit and integration tests.

•Project Veritas – Multi-Modal Misinformation Detection

February 2026

National 1st Place Winner, Echelon 2026

[GitHub](#)

- **Designed** a multi-modal fact-checking framework with four parallel inference streams (Visual, Semantic, Contextual, Audio) using PyTorch and Transformers, achieving **89.3% combined verdict accuracy** and **94.2% visual forensics accuracy** on the CIFAKE benchmark.
- **Integrated** LLM-powered explainability with Grad-CAM heatmaps and blockchain-based audit trails, producing structured, evidence-backed verdicts at **\$0.03 per analysis** – projected **\$108K/year savings** at 1M requests/month.
- **Shipped** a full-stack verification pipeline as a Chrome Extension for real-time social media fact-checking, served via FastAPI with **6.7 s end-to-end latency** on 3.2 GB GPU memory.

ACHIEVEMENTS

- 1st Place, PowerMind 2026:** Top rank among 50+ teams in a 24-hour AI and ML Hackathon organized by Adani Group.
- 1st Place, Echelon 2026:** 1st prize at a National-Level 24-hour AI and Cybersecurity Hackathon among 200+ participants.
- 1st Place, Genesis 2026:** 1st rank in a month-long AI, ML, and Web Development Hackathon.
- Kaggle Global Rank 4, WASP 2025:** 4th place internationally on the Kaggle leaderboard for the TRACS Shared Task.
- Amazon ML Summer School 2026 (MLSS):** Selected nationwide for Amazon's invitation-only ML program covering deep learning, probabilistic models, NLP, reinforcement learning, and causality taught by Amazon scientists.
- Finalist, ACM Summer Challenge 2025:** Top 2% in a Competitive Programming contest organized by ACM, NIT Surat Chapter.

LEADERSHIP AND EXTRACURRICULAR

- Secretary, Association for Computing Machinery (ACM):** NIT Surat Chapter – organized technical events and coordinated with 100+ student members.
- Core Committee Member, Mindbend 2026:** Gujarat's largest techno-managerial fest – managed logistics, finance, documentation, and sponsorship across 30+ events.